

HTTP TRAFFIC MONITORING AND INTRUSION DETECTION IN A VIRTUAL SOC ENVIRONMENT

Security Operation Center (SOC) Analysis Document

Role Focus:

SOC Analyst

Project Type:

Security Operations Center (SOC) Simulation Lab / Network Intrusion Detection Project

Tools:

Splunk (SIEM) | Snort (IDS)

Objective:

To build a virtual SOC environment that monitors HTTP traffic and analyzes web activity using Snort IDS and Splunk SIEM for network based detection and log correlation

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1. SCOPE

This lab covers:

- HTTP traffic generation in a controlled environment
- Intrusion detection using network-based IDS
- Log aggregation and analysis in a SIEM platform
- Detection of basic web request patterns (GET traffic)

2. SYSTEM ARCHITECTURE

Components:

- Web Server: Internet Information Services
- Intrusion Detection System: Snort
- SIEM Platform: Splunk

3. NETWORK DESIGN

- Windows Server → HTTP service (port 80)
- Ubuntu Server → Snort sensor monitoring traffic
- Attacker machine → HTTP request generation
- Splunk → centralized log analysis

Kali Linux is the attacker machine which is the external network (OPT) with IP 192.168.2.5 while windows server which is the victim in which web server is found has IP 192.168.1.1 (LAN Network)

4. IMPLEMENTATION

Web Server Deployment (Windows Server)

- Installed web server (IIS) via server manager

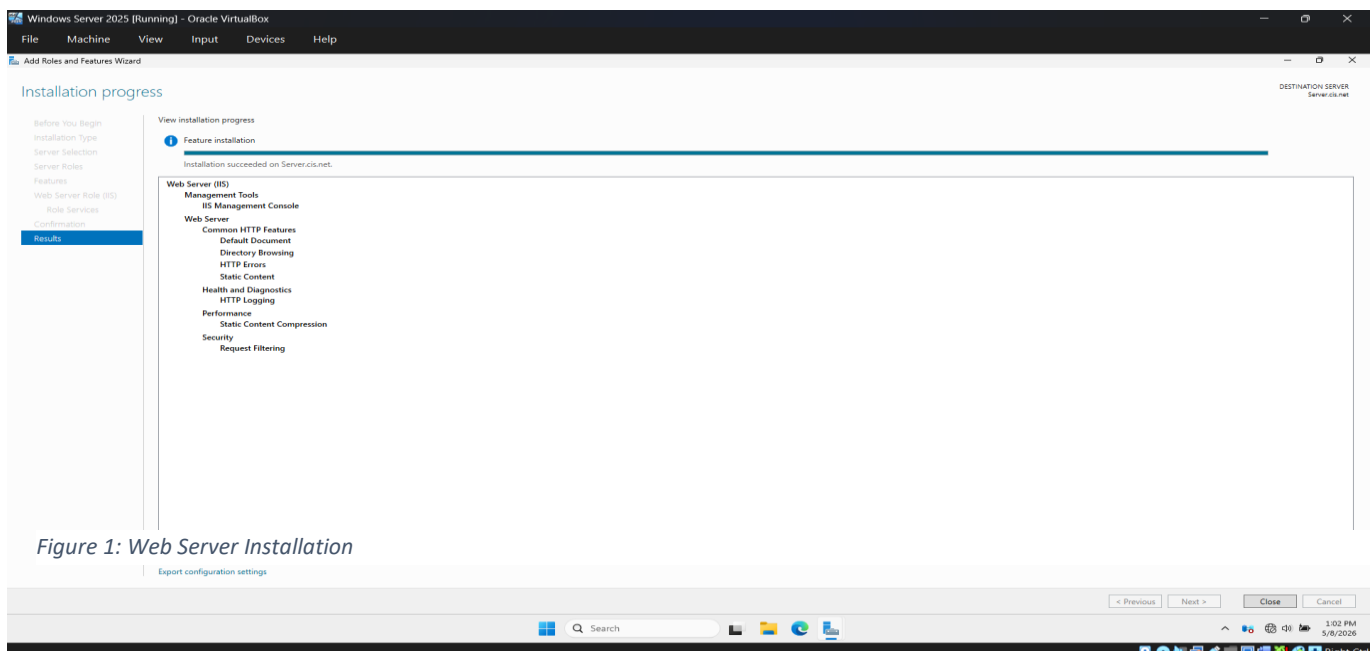
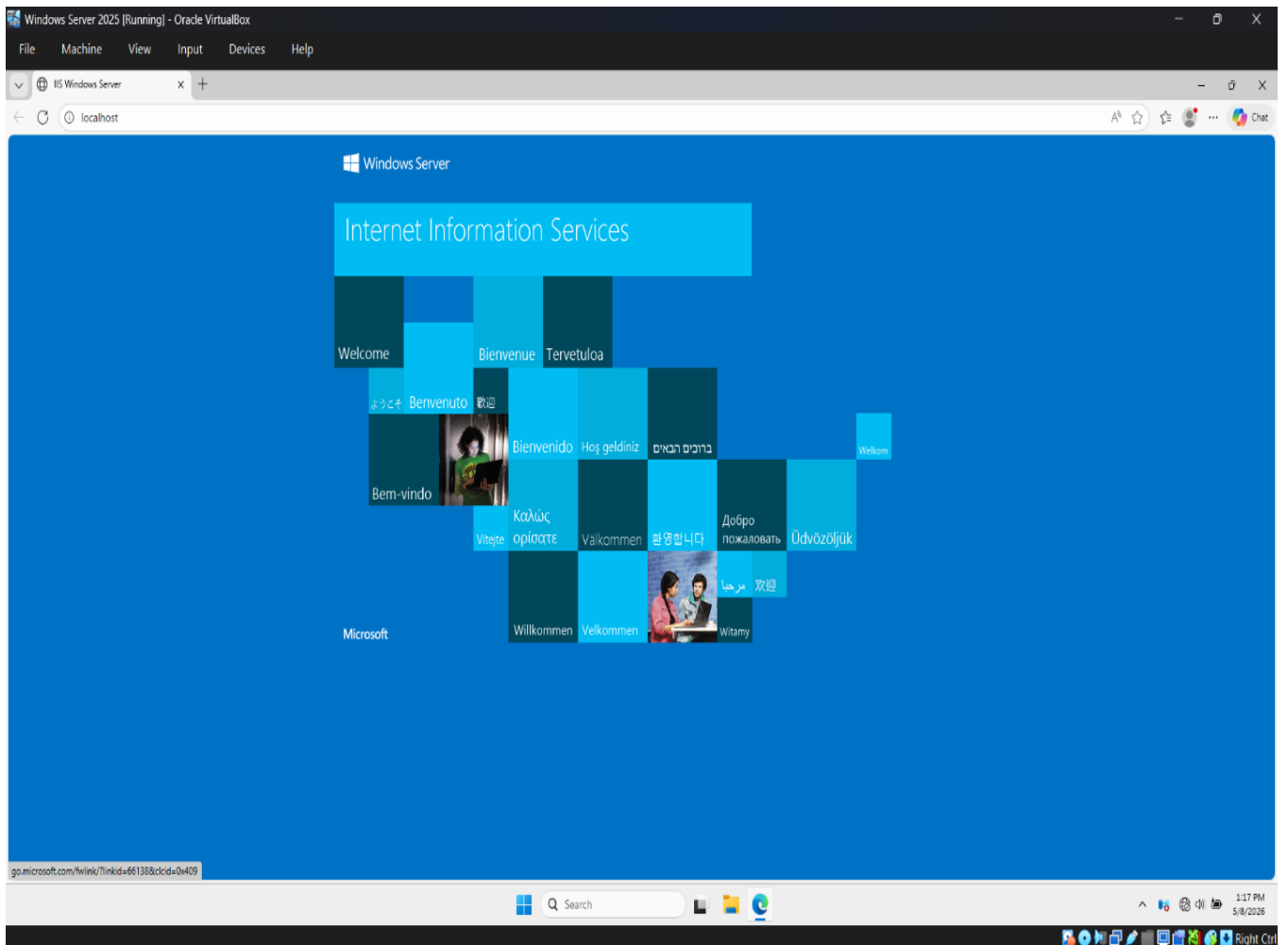


Figure 1: Web Server Installation

- Verified service availability on browser (<http://localhost>)



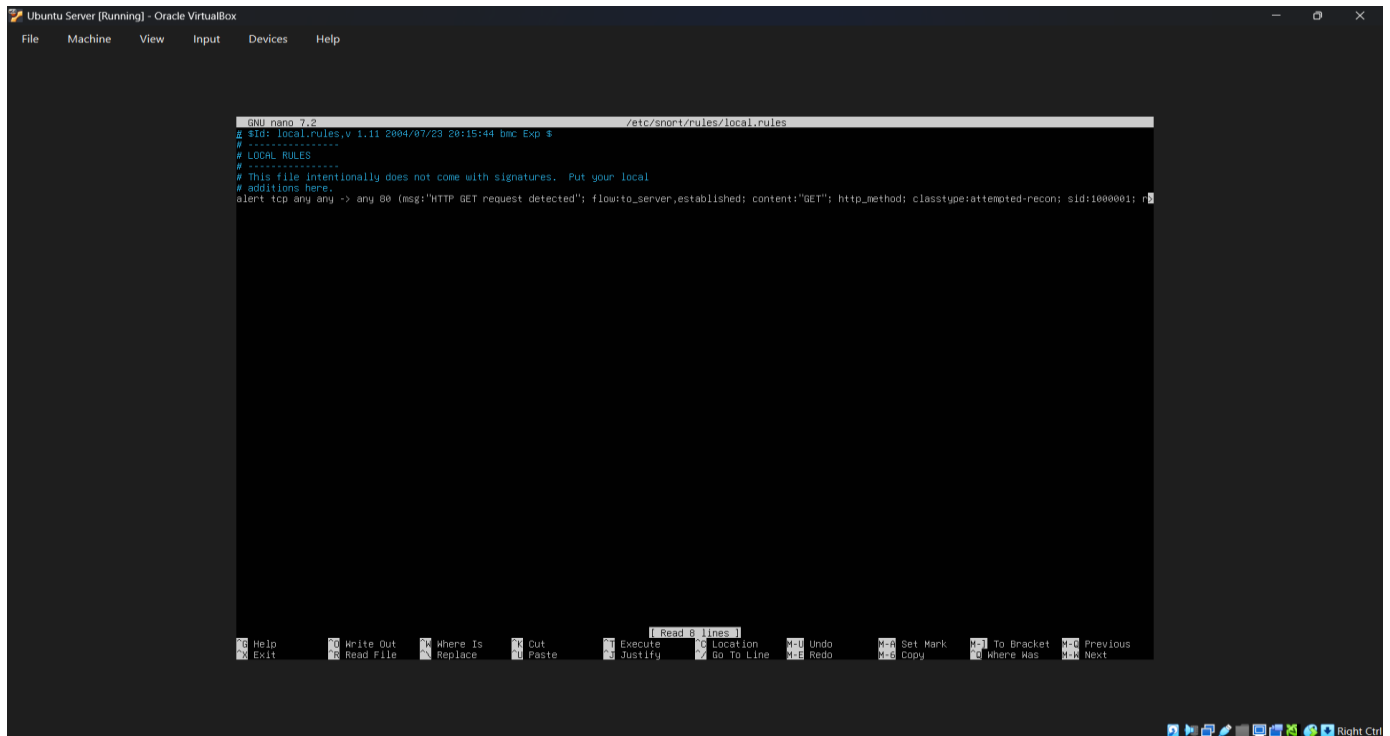
Intrusion Detection System Configuration (Snort)

Snort configured to monitor HTTP traffic on port 80

```

alert tcp any any -> any 80 (
  msg:"HTTP GET request detected";
  flow:to_server,established;
  content:"GET";
  http_method;
  sid:1000001;
  rev:1;)

```



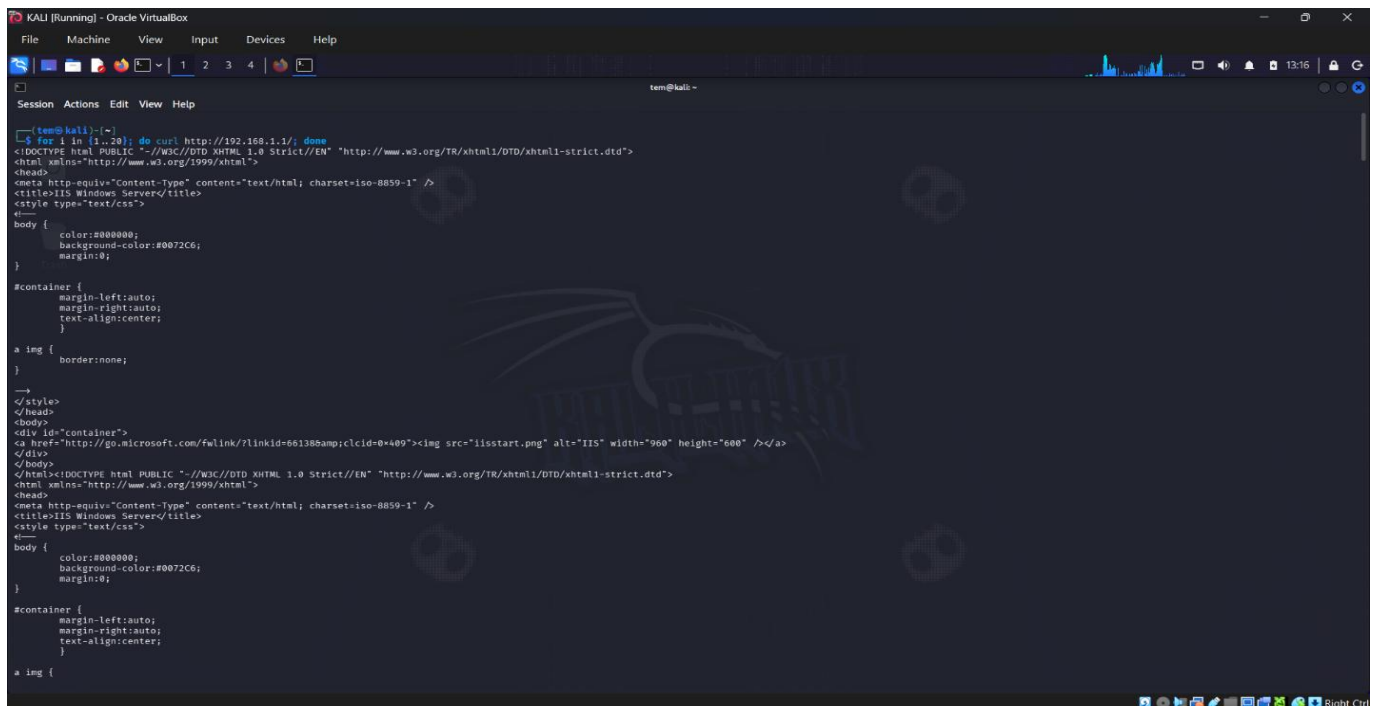
```
GNU nano 2.2 /etc/snort/rules/local.rules
# sid: local.rules.v 1.11 2004/07/23 20:15:44 hmc Exp #
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any 00 (msg:'HTTP GET request detected'; flowto_server,established; content:'GET'; http_method; classtype:attempted-recon; sid:1000001; rev:1)
```

SIEM Integration

- Snort logs forwarded to splunk Universal Forwarder
- Data indexed under (main)

5. TEST SCENARIO

- On kali Linux an attacker simulated to the web server

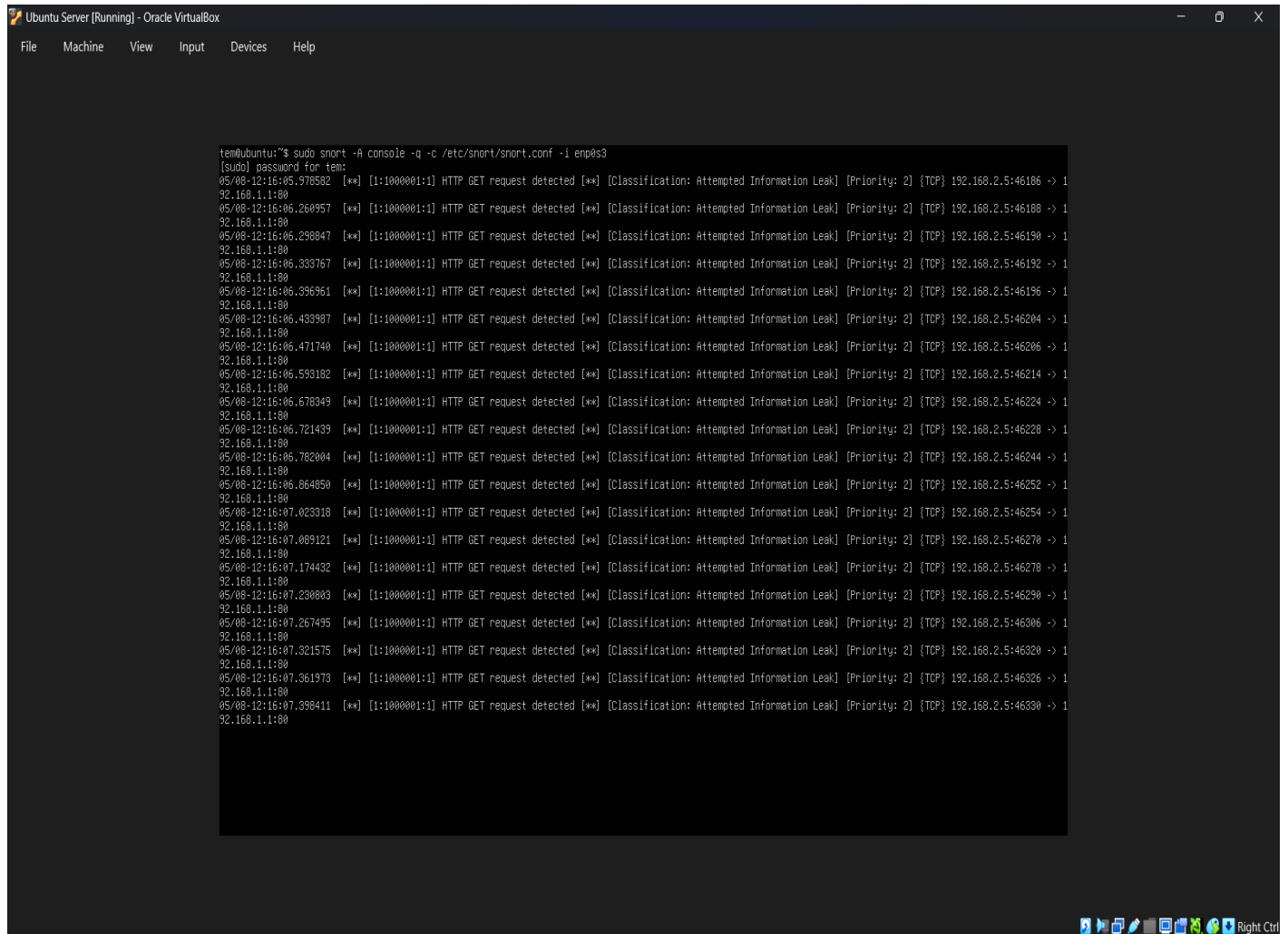


```
tem@kali: ~
$ curl http://102.168.1.1/.dowm
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" />
<title>IIS Windows Server</title>
<style type="text/css">
<!--
body {
    color:#000000;
    background-color:#0072C6;
    margin:0;
}
#container {
    margin-left:auto;
    margin-right:auto;
    text-align:center;
}
a img {
    border:none;
}
-->
</style>
</head>
<body>
<div id="container">
<a href="http://go.microsoft.com/fwlink/?linkid=66138&picid=0+409">img src="lisstart.png" alt="IIS" width="960" height="600" /></a>
</div>
</body>
</html><!--DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" />
<title>IIS Windows Server</title>
<style type="text/css">
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}
#container {
    margin-left:auto;
    margin-right:auto;
    text-align:center;
}
a img {
```

6. DETECTION RESULTS

Snort IDS Output

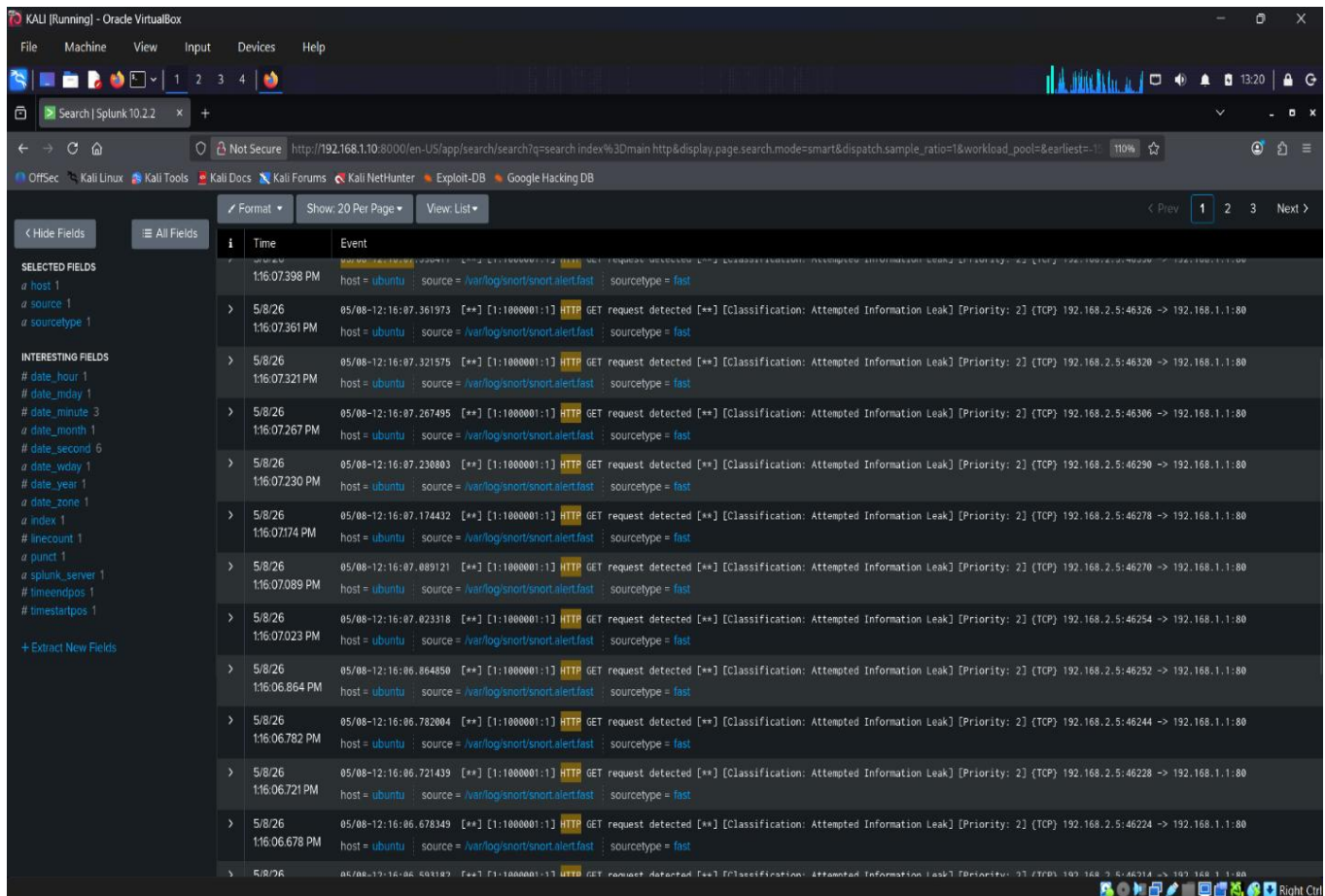
- Alerts triggered on HTTP GET request
- Each request logged as an individual event



```
tem@ubuntu:~$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
[sudo] password for tem:
05/08-12:16:05.978582  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46186 -> 192.168.1.1:80
05/08-12:16:06.269957  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46188 -> 192.168.1.1:80
05/08-12:16:06.298847  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46190 -> 192.168.1.1:80
05/08-12:16:06.333767  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46192 -> 192.168.1.1:80
05/08-12:16:06.396961  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46196 -> 192.168.1.1:80
05/08-12:16:06.439987  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46204 -> 192.168.1.1:80
05/08-12:16:06.471740  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46206 -> 192.168.1.1:80
05/08-12:16:06.593182  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46214 -> 192.168.1.1:80
05/08-12:16:06.678349  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46224 -> 192.168.1.1:80
05/08-12:16:06.721439  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46228 -> 192.168.1.1:80
05/08-12:16:06.782004  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46244 -> 192.168.1.1:80
05/08-12:16:06.864850  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46252 -> 192.168.1.1:80
05/08-12:16:07.029318  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46254 -> 192.168.1.1:80
05/08-12:16:07.089121  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46270 -> 192.168.1.1:80
05/08-12:16:07.174432  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46278 -> 192.168.1.1:80
05/08-12:16:07.230803  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46290 -> 192.168.1.1:80
05/08-12:16:07.267495  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46306 -> 192.168.1.1:80
05/08-12:16:07.321575  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46320 -> 192.168.1.1:80
05/08-12:16:07.361973  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46326 -> 192.168.1.1:80
05/08-12:16:07.398411  ** [1:1000001:1] HTTP GET request detected ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.5:46330 -> 192.168.1.1:80
```

SIEM Observation (Splunk)

- Increases in HTTP event frequency
- Source IP based traffic grouping
- Time series spike detection



7. ANALYSIS

- HTTP GET request are easily detectable at packet level
- Repeated requests from a single source indicate automated behavior
- Simple signature based detection generates high visibility but limited context

8. CONTROL FRAMEWORK MAPPING AND SECURITY INTERPRETATION

Identity (ID)

- Assets defined: web server, IDS sensor, SIEM platform

Protect (PR)

- Isolated lab environment ensures controlled traffic generation
- Segmentation between attacker and target systems

Detect (DE)

- Snort rules detect HTTP GET traffic
- Splunk identifies traffic patterns and spikes

Respond (RS)

- Alerts generated for HTTP activity events
- Logs centralized for incident review

9. LIMITATION

- Rule detects all HTTP GET requests (high false positive risk)
- No encryption analysis (no HTTPS visibility)
- No user authentication context
- Limited behavioral analytics

10. LESSON LEARNED

- Detection systems require context, not just signatures
- SIEM correlation is essential for meaningful analysis
- Lab accuracy depends more on architecture than rule complexity
- Baseline traffic understanding is critical before anomaly detection

11. CONCLUSION

This lab demonstrates the implementation of a basic SOC monitoring pipeline using IDS and SIEM tools to detect HTTP traffic. While effective for visibility, further tuning is required for real-world threat detection capability.